***Net revenues by product group***

| | Year Ended | | | % Variation | |
|---|---|---|---|---|---|
| | 2019 | 2018 | 2017 | 2019 vs 2018 | 2018 vs 2017 |
| | (In millions) | | | | |
| Automotive and Discrete Group (ADG) | $ 3,606 | $ 3,556 | $ 3,059 | 1.4% | 16.2% |
| Analog MEMS and Sensors Group (AMS) | 3,299 | 3,154 | 2,630 | 4.6 | 19.9 |
| Microcontrollers and Digital ICs Group (MDG) | 2,638 | 2,940 | 2,646 | (10.3) | 11.1 |
| Others | 13 | 14 | 12 | — | — |
| **Total consolidated net revenues** | **$ 9,556** | **$ 9,664** | **$ 8,347** | **(1.1)%** | **15.8%** |

For the full year 2019, our ADG revenues increased 1.4% compared to the previous period. The increase was primarily due to improved average selling prices of approximately 9%, which was entirely due to a better product mix, and partially offset by a decrease in volumes by approximately 8%. AMS revenues grew 4.6%, mainly due to the double-digits growth in Imaging. The increase was due to higher average selling prices of approximately 12%, as a result of a better product mix, and was partially offset by lower volumes of approximately 7%. MDG revenues were down by 10.3%, mainly due to Microcontrollers. The decrease was due to lower volumes of approximately 10% while average selling prices remained substantially flat.

In 2018, all product groups registered double-digit revenue increase. Our ADG revenues increased 16.2% for the full year 2018 compared to the full year 2017 on growth in both Power Discrete and Automotive. The increase was primarily due to improved average selling prices of approximately 21% and volumes decreased by approximately 5%. The increase in average selling prices was entirely due to improved product mix, while selling prices remained substantially flat. AMS revenues grew 19.9%, mainly on the strong increase in Imaging. The increase was due to higher volumes of approximately 12% and higher average selling prices of approximately 8%, which was entirely due to improved product mix of approximately 13%, while selling prices decreased by approximately 5%. MDG revenues were up by 11.1%, with Digital and Microcontrollers & Memories equally contributing. The increase was primarily due to higher average selling prices of approximately 11%, while volumes remained substantially flat. The increase in average selling prices was due to a better product mix of approximately 13%, while the selling prices effect was negative of approximately 2%.

***Net revenues by Market Channel***[1]

| | Year Ended December 31, | | |
|---|---|---|---|
| | 2019 | 2018 | 2017 |
| | (As percentage of net revenues) | | |
| OEM | 70% | 65% | 66% |
| Distribution | 30 | 35 | 34 |
| **Total** | **100%** | **100%** | **100%** |

(1) Original Equipment Manufacturers ("OEM") are the end-customers to which we provide direct marketing application engineering support, while Distribution customers refers to the distributors and representatives that we engage to distribute our products around the world.

Our revenues weight in Distribution registered a decrease of 5 percentage point compared to 2018, reaching a 30% share of total revenues in 2019. In 2018 as compared to 2017, our revenues weight in Distribution registered an increase of 1 percentage point.

***Net Revenues by Location of Shipment***[1]

| | Year Ended December 31, | | | % Variation | |
|---|---|---|---|---|---|
| | 2019 | 2018 | 2017 | 2019 vs 2018 | 2018 vs 2017 |
| | (In millions) | | | | |
| EMEA | $ 2,265 | $ 2,478 | $ 2,142 | (8.6)% | 15.7% |
| Americas | 1,351 | 1,264 | 1,085 | 6.8 | 16.5 |
| Asia Pacific | 5,940 | 5,922 | 5,120 | 0.3 | 15.7 |
| **Total** | **$ 9,556** | **$ 9,664** | **$ 8,347** | **(1.1)%** | **15.8%** |

(1) Net revenues by location of shipment are classified by location of customer invoiced or reclassified by shipment destination in line with customer demand. For example, products ordered by U.S.-based companies to be invoiced to Asia Pacific affiliates are classified as Asia Pacific revenues.

By location of shipment, in 2019, revenues grew 6.8% in Americas, driven by Power Discrete, remained substantially flat in Asia and decreased 8.6% in EMEA, mainly due to lower sales of Microcontrollers and Power Discrete. In 2018 revenues grew across all regions, led by Asia Pacific and EMEA, mainly due to growth in Imaging and Automotive.

***Gross profit***

| | Year Ended December 31, | | | Variation | |
|---|---|---|---|---|---|
| | 2019 | 2018 | 2017 | 2019 vs 2018 | 2018 vs 2017 |
| | | (In millions) | | | |
| Cost of sales | $ (5,860) | $ (5,803) | $ (5,075) | 1.0% | (14.3)% |
| Gross profit | $ 3,696 | $ 3,861 | $ 3,272 | (4.3)% | 18.0% |
| Gross margin (as percentage of net revenues) | 38.7% | 40.0% | 39.2% | -130 bps | +80 bps |

In 2019, gross margin decreased by 130 basis points to 38.7% from 40.0% in the full year 2018 mainly due to normal price pressure and increased unsaturation charges, partially offset by improved manufacturing efficiencies, better product mix, and favorable currency effects, net of hedging. Unused capacity charges in 2019 were $65 million, impacting full year gross margin by 70 basis points.

In 2018, gross margin improved by 80 basis points to 40.0% from 39.2% in the full year 2017 benefiting from manufacturing efficiencies and better product mix, partially offset by normal price pressure and unfavorable currency effects, net of hedging. In 2018 unused capacity charges were negligible.

***Operating expenses***

| | Year Ended December 31, | | | Variation | |
|---|---|---|---|---|---|
| | 2019 | 2018 | 2017 | 2019 vs 2018 | 2018 vs 2017 |
| | | (In millions) | | | |
| Selling, general and administrative expenses | $(1,093) | $(1,095) | $ (981) | 0.3% | (11.7)% |
| Research and development expenses | (1,498) | (1,398) | (1,296) | (7.1) | (7.9) |
| **Total operating expenses** | **$(2,591)** | **$(2,493)** | **$(2,277)** | **(3.9)%** | **(9.5)%** |
| As percentage of net revenues | (27.1)% | (25.8)% | (27.3)% | -130 bps | +150 bps |

The 2019 operating expenses increased 3.9% compared to the prior year, mainly due to salary dynamic, increased spending in certain R&D programs and higher share-based compensation cost, partially offset by favorable currency effects, net of hedging.

The 2018 operating expenses increased 9.5% compared to the prior year, mainly due to unfavorable currency effects, net of hedging, salary dynamic, increased R&D activities and higher costs of the share-based compensation plans.

The R&D expenses were net of research tax credits in France and Italy, which amounted to $126 million in 2019, $138 million in 2018 and $124 million in 2017.

***Other income and expenses, net***

| | Year Ended December 31, | | |
|---|---|---|---|
| | 2019 | 2018 | 2017 |
| | | (In millions) | |
| Research and development funding | $ 132 | $ 52 | $ 65 |
| Phase-out and start-up costs | (38) | (1) | (8) |
| Exchange gain (loss), net | — | 4 | 4 |
| Patent costs | (1) | (8) | (9) |
| Gain on sale of businesses and non-current assets | 7 | 8 | 4 |
| Other, net | 3 | (2) | (1) |
| **Other income and expenses, net** | **$ 103** | **$ 53** | **$ 55** |
| As percentage of net revenues | 1.1% | 0.5% | 0.7% |